

MA 1011A (Linear algebra and series)
2024–2025 (August–November Semester)

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Syllabus

- ▶ Systems of linear equations and their solutions. $\mathbb{R}^n$ and $\mathbb{C}^n$ as vector spaces and their linear subspaces; spanning set and linear independence; basis and dimension of a subspace of $\mathbb{R}^n$. Matrices and their associated four fundamental subspaces: null space, column space, row space of a matrix and null space of its transpose. Row rank and column rank.
- ▶ Elementary row operations on matrices and elementary matrices; Reduced row echelon form. Gaussian elimination method for solving systems of linear equations. Determination of row rank, column rank using elementary row operations; row rank equals column rank. Inverse and determinant of a square matrix. The Gauss-Jordan method of computing the inverse of an (invertible) matrix.
- ▶ Linear transformations and their matrix representations. Effect of a change of basis on their matrix representations. Null space and range of a linear transformation. Rank-nullity theorem.
- ▶ Eigenvectors and eigenvalues. Characteristic polynomials, Cayley-Hamilton theorem, eigenvalues of special matrices, algebraic and geometric multiplicity, diagonalization.
- ▶ Sequences and series of real numbers: tests for convergence.
- ▶ Power series, Taylor series and Fourier series.

References

- [1] S. Lang, *Introduction to linear algebra*, 2 ed., Springer, 2012.

Additional reading material

1. Linear algebra and its applications, Gilbert Strang, 4th Edition, Cengage Learning.